

Automated Essay Scoring for IELTS using Classification and Regression Models

Prithvi Sunilkumar Nair

Department of Networking and Communications School of Computing,
SRM Institute of Science and Technology, Kattankulathur,
Chennai, Tamilnadu, India
pn7190@srmist.edu.in

TYJ Naga Malleswari

Department of Networking and Communications
School of Computing
SRM Institute of Science and Technology
Kattankulathur, Chennai, Tamilnadu, India
nagamalt@srmist.edu.in

Abstract - The advancement of automated essay scoring (AES) mechanisms in the English proficiency evaluation through the standardized English proficiency tests like the International English Language Testing System (IELTS) is a hot subject these days. The purpose of this article, in particular, is to explore the application of regression and classification methods in both automated IELTS essay grading and customizing IELTS essay grades. Moreover, this study options classification and regression models to develop IELTS essay grading method by means of dataset of IELTS essays. The analysis employs methods to identify the linguistic and structural parts of essays, and also compares the model performance along with the baseline metrics. Automating the outcome provides a way for the grading process to eliminate human error and proves successful in assessing students' IELTS essays with accuracy. With this study, a path brings to the light new directions for the research of ways to increase the effectiveness and consistency of language proficiency ES exam ratings, and a way to develop automatic evaluation systems.

Keyword - Classification model, Regression model.

I. INTRODUCTION

Among the most frequently given ancient language scores to assess prospective applicants is IELTS or the International English Language Testing System. As a general rule, those IELTS writing assignments are manually and precisely graded by the skilled examiners, although this process is labor-intensive, subjective, and time-consuming. Over the years AES systems have proven to be a suitable mechanism to be utilized in the place of or alongside the human marking in the context of standardized exams in a bid to mitigate the arising difficulties. The study focuses on the concepts and accuracy of using classification and regression models for evaluation of IELTS essays using AI techniques. Our basic goal is, therefore, to develop a machine learning algorithm that reviews both clarity and quality of the language used in compositions responded to the questions posed during an IELTS test.

This research builds intensively on the wave of numerous studies having explored different methods and algorithms for automated essay scoring in diverse academic disciplines. In contrary the precise strategy that works for IELTS writing assignments which are based on the particular linguistic features and assessment criteria is

required, because the IELTS essays are varied in nature and scoring system.

Classification model:

A classification model is a kind of machine learning algorithm that is used to predict what class or category of labels an incoming data will fall into. This system learns by processing tagged and content related data, and then assembles this knowledge to categorize new and unmarked content. Algorithms of multiple types adopted over a wide scale: logistic regression, decision tree, support machines, and neural networks as well have been shown in the figure 1.

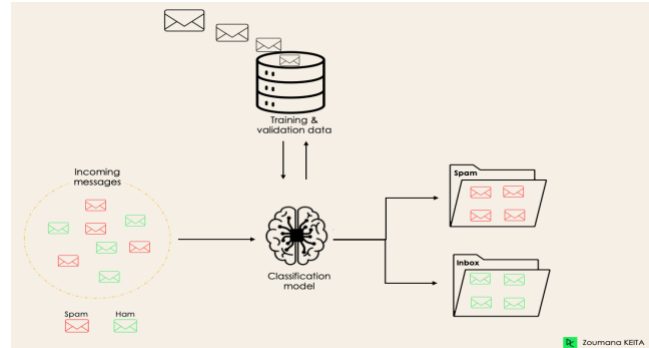


Fig. 1: Concept of a classification model

The accuracy, precision, recall, and F1-score measures are used to evaluate the model for the detection of a class, while the confusion matrix shows the dataset as a confusion matrix depicts the datasets as theirs.

Regression model:

Regression model is a notable type of machine learning technique which is applied for predicting continuously variable process results through using input data. The method works by collecting the data on the relationship of independent, deterministic factors, and the dependent one, learned by labeled datasets. It utilizes the comprehended knowledge and makes an inference in comparison to a new set of data. Regression models have many methods to choose from; namely: linear regression, polynomial regression, decision trees, random forest model, and neural networks.

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

Eq. 1: Regression model formula

Equation 1 depicts a linear interaction between the independent variables and the depended variable which are expressed as below where the explanatory variables are represented as coefficient and represent the influence of respective independent variable on the dependent variable. β_0 represent the independent variable. find out the best fit coefficients, which are those values that yield the minimum difference between the observed and predicted values of the dependent variable. The ultimate goal of regression analysis is finding out the coefficients' optimal values, one of many regression tasks. For the most part, the function is usually performed by such optimization methods as, e.g., the gradient descent or normal equation. We often measure goodness of fit/performance of regression models using mean squared error (MSE), root mean squared error (RMSE), and R-squared. All these evaluate the accuracy and ability in capturing the real intent of the data present.

This paper is an investigation that explores the applications of regression and classification models for systemizing essay rating in the circumstances of the academic IELTS institution. This study outlines our approach, which comprises the following steps: picking different parameters of essay taking and many classification and regression algorithms for application, as well as their evaluation based on predefined metrics. The present research intends to draw a pathogram and therefore, proposes how to design and implement an assessing procedure for language proficiency testing while classifying the issues with automated essay grading for IELTS. Now, the development of an effective automated scoring system with very good reliability and accuracy can help to make the essay grading process larger in scale, objective and efficient not only for the benefit of the test takers but also for educational institutions across the globe.

II. LITERATURE REVIEW

[1] This project aims to improve the content of essays for higher education students through text analytics metrics to provide instrumental advice on student writing. By the process of literature through the Dutch language corpus and the neural networks learning Levenberg-Marquardt algorithm and back elimination, scientists successfully extracted many of the features. The research also compared the models with relation to accuracy of evaluation of student papers in Dutch. The judgment of participant marking of

was evaluated using Cohen's kappa coefficient. We used the backward elimination technique to reduce the number of explanatory variables applied to regional reliability assessment. This approach is findings a data-driven approach which can be used to offer less information to automatic feedback of the native Dutch students' essays.

[2] There're efforts to develop AI-assisted system for automatic essay evaluation with the help of the most advanced NLP and graph methodologies. The system will appraise and review papers accomplished by value and the information generated by the speed of writing with disregard of the similarity of sentences. The deep learning models such as Deep Neural Nets (DNNs) and the Recurrent Neural Nets (RNNs) tends to find its application in different areas ranging from computer vision to natural language processing. Although, these models take into consideration other factors too but their performance gets a bit of drastic outcome. The system employs modern graph-based functionality to augment margins of its process. The most recent versions of algorithms such as Random Forest Regressor and Deep Neural Networks are used along with large efforts to clean data to refine the outcomes(improve the accuracy). The system employs Google's says word2vec model and graphs as the extractors of features. The platform enables thorough rating of the essays and it provides the essay authors with feedback and recommendations; however, there is still a lot of unexploited possibilities that could be used for the development of the system.

For instance,[3] a study conducted by an expert showed that a time efficient essay scoring technique to be used by the English teachers so that they would overcome the challenge of time constraints when grading and offering feedback is what this article entails. The process as defined through the means of a Kaggle dataset that contains 3000 features, arranged according to the term frequency principle with an additional manual set of 10 features serves as a criterion for the evaluation stage. At its core is the use of machine language techniques characterized by NLP [14] that transforms the text into vectors fitted with 3-tiered features. The study involved 3911 pieces of writing and applied machine learning techniques. The study came up with the conclusion that XGBoost Regression with GridSearchCV was the most exact model among all others. The major polynomial characteristics immediately stand out as the number of words, number of sentences, number of lemmas, number of nouns, and number of verbs.

[4] This study is aimed at supplying a method of sentence grade evaluated student essays which are empirically accessible when classifying sentence-level datasets. Two models were created: the first adopting a pre-trained model and the other created by a neural network based on multi-head-attention method. The model is composed of three components: a model which is pre-trained, an embeddings dimension reducer and an outcome assessor. The sentence-based model which is highly efficient for processing of data properly addresses outliers,

displays high robustness punctuation marks, maintaining the best possible order of numbers and generations of contractions in their natural form thus keeping the structure of the sentence intact. The sentence neighboring word-level model builds a sentence tokenized version of this one and a POS entries tag to label the token sequence. The authors gathered a brand fresh dataset with 400 essays and two different structures of neural network in order to touch on the problem of a persuasive student essay evaluation.

[5] The paper starts with the concept of a well advanced automatic English system scoring (AES) which runs Random Forest algorithm (RF) to get the most accurate essay scores. The system is organized into modules for end user login, evaluation, statistic (sic figure), and error correction. RF, GBDT, and XGBoost are examples of the models chosen for this task to make the prediction of scores of English essays. The model achieved this by merging separately preceded featuring techniques – Random Forest, Gradient Boosting Decision Trees and Extreme Gradient Boosting. The RF model shows a significant elevation in feature selection indexes of a combined factor which brings with it withholding degree attribute along with non-text features. In case of this RF model the results for the ASAP dataset are better in comparison with the test data and critic.com.

[6] used ChatGPT [17], an advanced AI language model to not only conduct automated essay scoring (AES) on a large data set of 12,100 essays found on the TOEFL Corpus of Non-Native Written English [TOEFL11]. The GPT indicates that is it high precision and safety level that makes it a great assistant to current human assessment skills. The present investigation postulates that the modern AI writing accentuation and correction instruments can be the next big revolution in the field of writing evaluation and feedback [13] in research and practice, which might pave the way for the transformation of many sectors. In addition, the research should also emphasis on sharpening and integration of linguistic tackles to achieve high rate of effectiveness.

[7] In this article, we introduce a method of incorporating classification technique in the analysis with different Automated Essay Scoring Systems (AES) including their scorings patterns. The process involves three main steps: the network-based architecture or tree-shaped model is trained, item response theory (IRT) parameters are estimated, and integrated scores are predicted. The method of GMFRM tests yields a noticeably high overall accuracy than the other methods, which implies that every attributes of the rater is beneficial to the improvement in accuracy. The method has power to provide with rounded grades unlike those scores which some methods present.

[8] The authored Smart Innovative LSTM semantic model for measuring English essay that produces its output by detecting the most significant semantic scoring traits is described in the article. The elevated XGBoost algorithm shows its superiority over the conventional XGBoost and LSTM substantially thereby sniffing the highest kappa value

that is used to measure the level of English essay. The XGBoost algorithm takes into account various features in order to carry out the analysis of the structure, degree of compatibility, and content composition of the written samples. It is the holistic approach to the scoring process that allows for precision in calculation as well as speed when handling essays of different lengths.

On the basis of reliable source [9] The system combines the latest technology of the natural language process to evaluate the English compositions which leads to unbiased evaluation and resultant can easily correlate to the given topics or question. The solution contains developed Python 3.8 and Django 3.0 with WordNet understanding address semantic ambiguity. The system could reached impressibility high accuracy rate of 89.1% during training and the speed level of 85 % approximately during the whole test. With the use of Grammatomat, those grammatical mistakes are allowed because of a multinomial HMM, Gaussian Mixture Model, and Bidirectional Neural Network. The system is like that and it helps the evaluations of essays in several languages, for instance, Tamil, Telugu and Malayalam. Therefore, it is worth in school settings.

III. MOTIVATION

Achieving a desired result and getting higher grades can be accomplished if effective free resources that enable one to sharpen his essay writing skills are available. Therefore, the establishment of a free service for IELTS writing tasks sample preparing students gives small leap of their writing skill, gets reviews, and raises an opportunity of passing the exam, consequently, which provides education and career goals.

IV. ARCHITECTURE

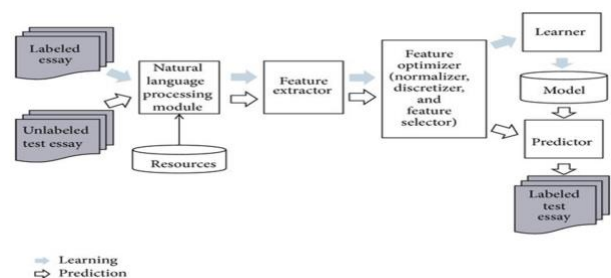


Fig. 2. Basic Architecture of the system

After a user or student type in their essay whatever it is task 1 or not labeled task 2, the instant alert to the nlp module! The data set is pre-processed to detect and remove noise, finally it is divided into train and test dataset before going to the feature extraction stage. At this stage certain features are selected from the dataset. Say, we assume that the length of the article also becomes a key feature of the model as well as those compiled that will help in predicting the scores. In order to facilitate the model to learn the scope of this data, we select only the most appropriate features and provide it to the model that is constructed. The teaching of this skill is

focused not just on prediction but on problem identification and solving as depicted in figure 2.

V. IMPLEMENTATION

However, step one to set a ball rolling was data collection and to infer the graded IELTS writing test dataset from Kaggle. Data retrieval with prior preprocessing operations followed in the line of writing. Through the use of few symbols (punctuation), and the replacing of numbers with values, the informational content of the text was preserved. At the same time, the word count for each essay was also set as another of the attributes being studied through the length of the essay.

Exploratory Data Analysis (EDA) helped a better understater of the features of the dataset. The use of the histogram-based graphs helped a lot for the clarification of the distribution pattern of paper lengths and IELTS scores. As a third step, there was a study which illustrated the correlations that the articles of the essay had with the scores of the IELTS, as shown in Figure 3.

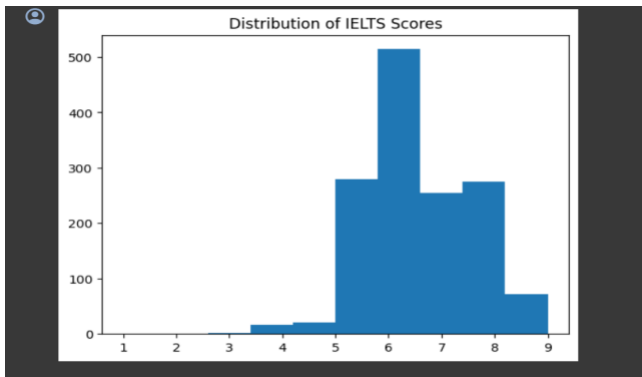


Fig. 3. Bar graph comparing IELTS score with length of the essay

The key point here is to develop the task variety in such a way that it could be carried out by more autonomous neural networks by means of feature engineering. The Text Vectorization layer of TensorFlow was used for the process of tokening and vectoring textual data. More than that, it is also based on previous pre-trained models like Word2Vec and USE embeddings, which are intended for understanding the diverse, symbolic meanings.

Therefore, classifiers and regression models were next build. A compound model which took care of both pre-trained embeddings, i.e., Universal Sentence Encoder (USE), and Multilayer Perceptron (MLP) was devised for classification objectives. To make a model for prediction of IELTS scores with help of text data, the long-short term memory (LSTM) [2][12] based recurrent neural network (RNN) model has also been established. On the other side, USE model was used for re-shaping and improving its classifications capabilities. On the subject of the regression part of the accurate IELTS scores prediction, the hybrid model with multi-layer perceptron (MLP) and the trained

USE was presented. A new model was developed (only for the post-training USE regression) so as to capture some of the intricacies and limitations associated with certain types of adjustments (the very nature of the in-use condition itself). The model conducts comparison with a hybrid regression that is made up of MLP with the Word2Vec embedding [2].

This included, carefulness in the training and modelling assessments was put to good use. Tracking of the model accuracy throughout training was done with optimizers and loss functions from the suitable set being utilized. For the purpose of the model capacity assessment generalization, validation datasets have been used. The feature like as accuracy, loss and RMSE were critical parameters we used to judge the model efficacy. At the validation stage, numerical criteria were used to determine differences between various models.

The predictive power of the model that performed nicely in regression analysis was assessed through the utilization of random prediction analyses. To check the reliability of the constructed model, we brought excels by predicting the IELTS scores and comparing the results with true scores in the test set.

The last decision to make the final model was based on the validation results, considering several criteria: predictive power, accuracy, and the RMSE. Whenever the dataset contains these flaws, such as imbalance or other biases, these problems have been properly addressed. In light of the reviews of the study, research agenda and possible further investigate areas were shared as a next step.

VI. RESULTS

The initiative that the study undertook to create neural network models for automated essay scoring that incorporate Task_Type, sentence embeddings, and word embeddings has proven to be extremely enlightening. In spite of the fact that the classification models produced dismal results with an accuracy of only 0.27.

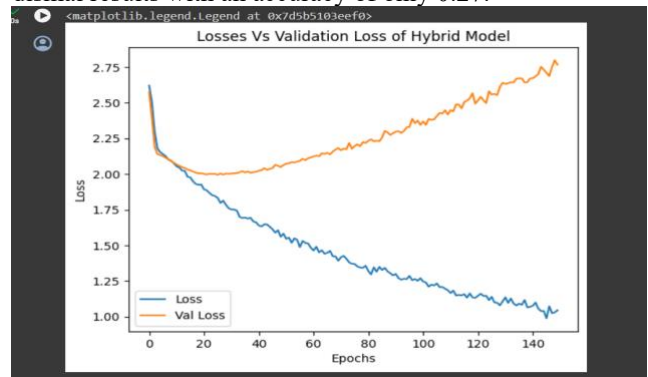


Fig. 4. Losses Vs Validation loss of hybrid model

The Figure 4 compares and shows us the deviation of the losses and the validation loss for the Hybrid model made using the mlp and use model and it can be seen that the

validation loss falls down first and then keeps gradually increasing after the point of 2.10 while the losses keep decreasing gradually.

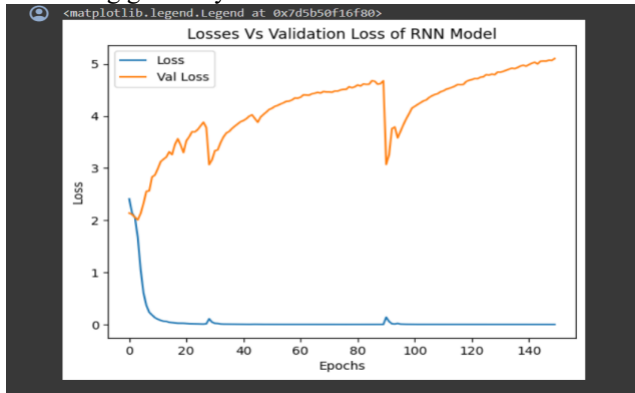


Fig. 5. Losses Vs Validation loss of RNN model

The Figure 5 compares and shows us the deviation of the losses and the validation loss for the RNN model [2][12] and it can be seen that the validation loss starts at around 2 increases to a point above 4, fluctuates again near 3 and increases gradually while the losses falls down straight and remains almost constant after that.

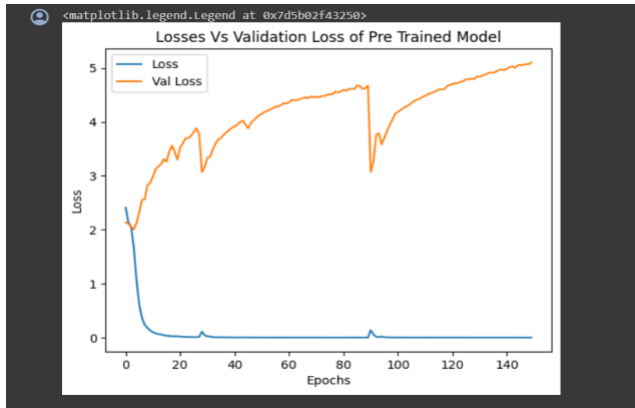


Fig. 6. Losses Vs Validation loss of pre trained model

The Figure 6 compares and shows us the deviation of the losses and the validation loss for the pre trained and use model and it can be seen that the validation loss fluctuates a lot but increasing over time with epochs while the losses again fall sharply and then remain almost constant. Our investigation into regression models showed that they were more promising. This study was able to successfully minimize the validation loss (mean squared error) to a noteworthy 0.88 by utilizing regression approaches as shown in Figure 7. This is an indication of the model's capability to predict scores with a margin of around one point.

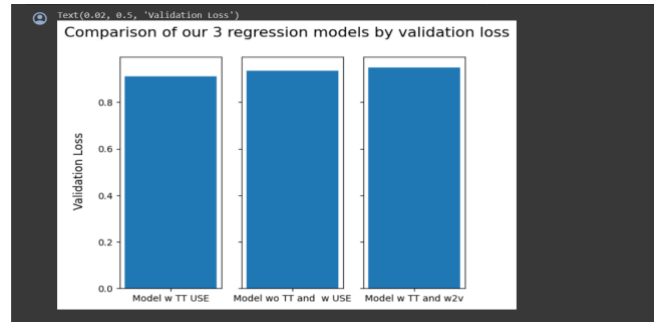


Fig. 7. Comparison of three regression model by validation loss

It can be seen in Figure 7 this comparison bar graph between all the three regression models that this study has managed to cut the validation loss by a decent margin. Usually these models have a validation loss around 1 so a score around 0.88 is considerably a good achievement. Notably, the fact that such a low validation loss was achieved with the quantity of the dataset that is now available highlights the possibility of further improvements brought about by extended data. There is a convincing possibility that the performance of the model could be improved with the addition of additional data. In conclusion, our findings indicate that classification models might not be enough for this complicated task; rather, regression models present a feasible option for the improvement of automated essay scoring systems.

VII. CONCLUSION

To wrap up this study, we designed and tested a set of neural network models to automate the essay scoring process. Exploration of the major successes included the application of diverse experimental techniques as well as the investigation. This project was commemorated with a neural network architecture that unified both task type and text embeddings. Such models have done quite well both in the classification & regression tasks which result in an ability to link essay features to IELTS scores. USE and word2vec embeddings enhance the prediction accuracy of the model by retaining the semantic information and representing the text features. The classifiers side has been beaten by regression in predicting overall IELTS scores. First example will be the comparison highlighted with the model selection and engineering feature importance for the performance. On the other hand, windows became clear positive side and areas of improvement seemed as clear. The imbalanced dataset alongside the small amount of training data made it difficult to start the models and get generalization right. Pre-trained word embeddings may distort the lifeworld details of the domain. We could make the models able to do the scoring for IELTS writing essays automatically, but we need to do more work and overcome the restrictions to obtain a general model that fully reflects all languages and topics. This article envisages neural networks for reading

educational process marks and magical auto Scrutiny Systems for Language Proficiency Tests.

REFERENCES

- [1] M. Abbas, P. van Rosmalen and M. Kalz, "A Data-Driven Approach for the Identification of Features for Automated Feedback on Academic Essays," in *IEEE Transactions on Learning Technologies*, vol. 16, no. 6, pp. 914-925, Dec. 2023, doi: 10.1109/TLT.2023.3320877.
- [2] V. Suresh, R. Agasthiya, J. Ajay, A. A. Gold and D. Chandru, "AI based Automated Essay Grading System using NLP," *2023 7th International Conference on Intelligent Computing and Control Systems (ICICCS)*, Madurai, India, 2023, pp. 547-552, doi: 10.1109/ICICCS56967.2023.10142822.
- [3] Shutong Yang, "Automated English essay scoring based on machine learning algorithms," *2023 2nd International Conference on Data Analytics, Computing and Artificial Intelligence (ICDACAI)*, pp. 263-266, 2023.
- [4] M. Hossain and H. A. Mustafa, "Automated Writing Evaluation Using Sentence by Sentence Scoring Model," *2023 International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM)*, Gazipur, Bangladesh, 2023, pp. 1-6.
- [5] L. Wu, "Automatic English Essay Scoring Algorithm Based on Machine Learning," *2023 IEEE International Conference on Integrated Circuits and Communication Systems (ICICACS)*, Raichur, India, 2023, pp. 1-4.
- [6] Mizumoto, A., & Eguchi, M. (2023). Exploring the potential of using an AI language model for automated essay scoring. *Research Methods in Applied Linguistics*, 2(2), Article 100050. <https://doi.org/10.1016/j.rmal.2023.100050>
- [7] Research Methods in Applied Linguistics, vol. 2, no. 2, p. 100050, Aug. 2023, doi: 10.1016/j.rmal.2023.100050.
- [8] "Integration of prediction scores from various automated essay scoring models using item response theory," *IEEE Journals & Magazine | IEEE Xplore*, Dec.01,2023. <https://ieeexplore.ieee.org/document/10061235>
- [9] "Machine Learning Models: A study of English essay text content feature extraction and automatic scoring," *River Publishers Journals & Magazine | IEEE Xplore*, Nov. 01, 2023. <https://ieeexplore.ieee.org/document/10326100>
- [10] S. Rajest, R. Rajan, S. T, and S. R, "A new Natural Language Processing-Based Essay Grading algorithm," *ResearchGate*, May 2023, [Online]. Available: https://www.researchgate.net/publication/370772389_A_New_Natural_Language_Processing-Based_Essay_Grading_Algorithm
- [11] I. Persing and V. Ng, "Modeling argument strength in student essays," in *Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics and the 7th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, 2015, pp. 543–552
- [12] MacCartney, B., & Manning, C. D. (2011). Automated evaluation of short answer quality. *Computational Linguistics*, 37(2), 279-318.
- [13] Palanikkumar, D., Abhisek Sethy, Kasiprasad Manneppalli, P. John Augustine, Semira Abdela Seyid, and Adem Aman Hassen. "Managing the Offline Text Recognition Accuracy Using Recurrent Neural Network." *Journal of Contemporary Issues in Business and Government* Vol 27, no. 1 (2021).
- [14] P.Chinnasamy V.Suresh K.Ramprathap B.Jency A.Jebamani K.Srinivas Rao M.Shiva Kranthi "COVID-19 vaccine sentiment analysis using public opinions on Twitter" *Materials Today: Proceedings* Volume 64, Part 1, April 2022, Pages 448-451, ISSN: 2214-7853
- [15] F. Dong, Y. Zhang, and J. Yang, "Attention-based recurrent convolutional neural network for automatic essay scoring," in *Proceedings of the 21st conference on computational natural language learning (CoNLL 2017)*, 2017, pp. 153–162.
- [16] M. Cozma, A. M. Butnaru, and R. T. Ionescu, "Automated essay scoring with string kernels and word embeddings," *arXiv preprint arXiv:1804.07954*, 2018.
- [17] M. Zaheer, G. Guruganesh, K. A. Dubey, J. Ainslie, C. Alberti, S. Ontanon, P. Pham, A. Ravula, Q. Wang, L. Yang et al., "Big bird: Transformers for longer sequences," *Advances in Neural Information Processing Systems*, vol. 33, pp. 17 283–17 297, 2020.
- [18] P. He, X. Liu, J. Gao, and W. Chen, "Deberta: Decoding-enhanced bert with disentangled attention," *arXiv preprint arXiv:2006.03654*, 2020.